E-score

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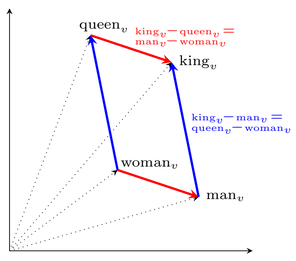
Abstract

- Sequence similarity is central in biology, as similar sequences share function and common ancestry. BLOSUM matrices play a crucial role in identifying similarities, but have the drawback that they are independent of context.
- We propose a new method, ***E*-score**, for amino acid similarity that uses the context of the protein sequences. It uses the cosine similarity of the embedding vectors associated with the residues.
- E-scores greatly surpass the ability of BLOSUM matrices to produce true alignments.
- The new method can be applied to the similarity of any textual data, with many applications.

Embeddings

Natural Language Processing embeddings

- Associate a vector with each word by self-supervised learning on large unlabelled data (Common Crawl, Wikipedia, etc.)
- Context independent: word2vec,[6] GloVe
  - Capture meaning: $king_v - man_v + woman_v \approx queen_v$.
- Contextual: ELMo, BERT,[3] RoBERTa, XLNet, T5[7]
- The most successful ones rely on the transformer architecture

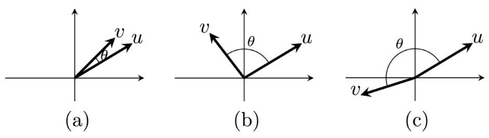


Protein embeddings

- Associate a vector with each amino acid residue; trained on protein sequences from UniProt
- Examples: ProtVec, SeqVec, PRoBERTa, MSA-transformer, ESM-1b and ESM2,[8] ProtTXL, ProtBert, ProtXLNet, ProtAlbert, ProtElectra, and ProtT5[4]

E-score (see full paper[1] for details)

The *cosine similarity* between two vectors u and v is the cosine of the angle, θ, between them; $\cos(\theta) \approx 1$ for similar (a), $\cos(\theta) \approx 0$ for orthogonal (independent) (b), $\cos(\theta) \approx -1$ for opposite (c).



For $u = (u_i)_{i=1..n}, v = (v_i)_{i=1..n}$, we have:

$$\mathrm{CosSim}(u,v) = \cos(\theta) = \frac{u \cdot v}{\|u\|\|v\|} = \frac{\sum_{i=1}^{n} u_i v_i}{\sqrt{\sum_{i=1}^{n} u_i^2}\sqrt{\sum_{i=1}^{n} v_i^2}}\,.$$

E-score definition

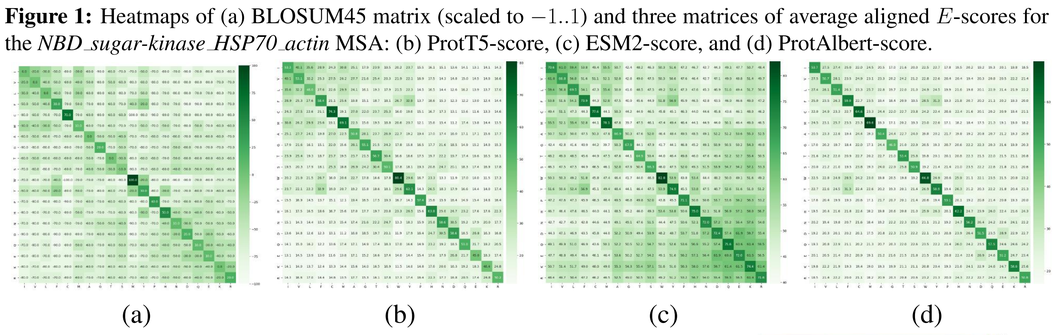
Given an embedding method E, two proteins P and Q, and two positions, i within P and j within Q, the *E-score* between the residues $P[i]$ and $Q[j]$ is defined as the cosine similarity between the corresponding embedding vectors, $E(P)$ and $E(Q)$.:

$$E\text{-score}(i,j) = \mathrm{CosSim}(E(P)_i, E(Q)_j)\,.$$

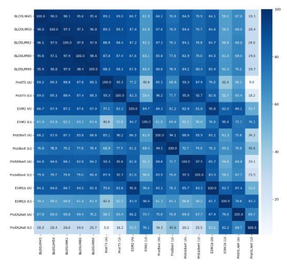
E-score example matrices

While a BLOSUM matrix is fixed, the E-scores change with context, so we computed average scores for each pair of amino acids. We distinguish between "aligned" (A) residues, which appear in the same column of the MSA, and "unaligned" (U) residues, which occur in different columns.

Figure 1: Heatmaps of (a) BLOSUM45 matrix (scaled to $-1..1$) and three matrices of average aligned E-scores for the *NBD_sugar-kinase_HSP70_actin* MSA: (b) ProtT5-score, (c) ESM2-score, and (d) ProtAlbert-score.



Pearson correlations between five BLOSUM matrices and twelve E-score matrices. ProtT5 matrices have the highest correlation with all BLOSUM matrices, followed, in order, by ESM2 and ProtBert, and then ProtAlbert, ESM-1b and, by far the last, ProtXLNet. For each method, the aligned matrix has higher correlation with BLOSUM matrices than the unaligned one.



Availability

The program to compute alignments based on various E-scores is available as a web server at `e-score.csd.uwo.ca`. The source code is freely available for download from `github.com/lucian-ilie/E-score`.

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ProtT5-score outperforms BLOSUM45

Experimental setup

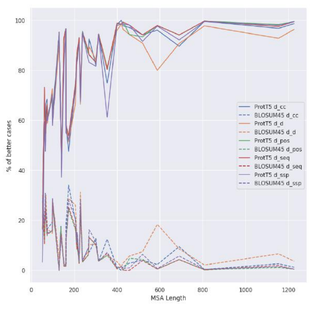
- Reference MSAs: wide variety from NCBI's CDD (Conserved Domain Database)[5]
- Alignment distances used to measure how close the ProtT5-score or BLOSUM45 dynamic programming alignments are to the reference: $d_{pos}, d_{seq}, d_{ssp}$,[2] and d_d, d_{cc} (new)
- 1000 pairs randomly chosen from each MSA; average distance to reference computed

Table 1: ProtT5-score against BLOSUM45 matrix: comparison using MSAs from Conserved Domain Database. The best values are shown in boldface. We show only the distances d_{cc}, d_d, and d_{pos}; the comparisons using d_{ssp} and d_{seq} are similar to the one for d_{pos}. Wilcoxon signed-rank test P-values are also shown.

MSA	d_{cc}			d_d			d_{pos}		
Conserved domain	ProtT5	BLOSUM45	P-value	ProtT5	BLOSUM45	P-value	ProtT5	BLOSUM45	P-value
Bbox2_TRIM37_C-VIII	0.0043	**0.0034**	2.4E-05	0.0013	**0.0010**	1.9E-07	0.0508	**0.0429**	4.9E-03
CBD_like	**0.0108**	0.0206	5.2E-67	**0.0037**	0.0100	9.3E-85	**0.1518**	0.2315	2.3E-82
Bbox2	**0.0117**	0.0249	1.6E-38	**0.0031**	0.0067	2.7E-34	**0.1361**	0.1910	8.0E-39
ChtBD1	**0.0127**	0.0221	2.0E-23	**0.0034**	0.0082	7.1E-25	**0.1609**	0.2200	5.0E-23
Bbox1_CYLD	**0.0137**	0.0284	1.8E-25	**0.0045**	0.0073	3.1E-09	**0.1553**	0.2211	1.4E-21
KAZAL_FS	**0.0133**	0.0391	8.1E-105	**0.0033**	0.0139	1.2E-100	**0.1470**	0.2699	1.1E-94
bHLH_SF	**0.0072**	0.0289	6.2E-83	**0.0029**	0.0133	2.1E-78	**0.0999**	0.2177	2.1E-81
CD_CSD	**0.0092**	0.0233	1.5E-79	**0.0015**	0.0061	2.9E-106	**0.1077**	0.1980	9.3E-93
C1	**0.0133**	0.0221	1.1E-58	**0.0038**	0.0083	4.9E-49	**0.1615**	0.2218	6.0E-48
TrHb	**0.0045**	0.0416	8.9E-05	**0.0010**	0.0077	1.4E-04	**0.0941**	0.3460	8.9E-05
Hb	**0.0015**	0.0025	5.1E-06	**0.0002**	0.0007	9.4E-09	**0.0441**	0.0728	4.6E-08
Hb-beta-like	**0.0012**	0.0020	2.0E-10	**0.0002**	0.0006	5.5E-16	**0.0227**	0.0390	5.6E-15
GS_GGDEF_2	**0.0018**	0.0120	1.7E-45	**0.0003**	0.0034	5.5E-44	**0.0272**	0.1665	8.0E-48
Globin-like	**0.0324**	0.1605	6.3E-21	**0.0065**	0.1186	5.4E-21	**0.3702**	0.8069	6.0E-21
FHb-globin	**0.0012**	0.0022	2.1E-46	**0.0001**	0.0006	2.8E-87	**0.0240**	0.0423	1.2E-78
PFM_HFR-2-like	**0.0069**	0.0093	1.5E-13	**0.0016**	0.0021	6.8E-26	**0.1043**	0.1396	1.1E-43
SH2_STAT_family	**0.0206**	0.0376	1.3E-95	**0.0066**	0.0146	2.0E-72	**0.2116**	0.3059	1.6E-109
PBP-like	**0.0127**	0.0388	3.7E-62	**0.0036**	0.0167	2.3E-53	**0.1983**	0.3531	5.2E-64
SH2	**0.0342**	0.0693	1.2E-121	**0.0090**	0.0289	4.4E-113	**0.3024**	0.4788	4.1E-137
Globin_sensor	**0.0045**	0.0236	2.7E-149	**0.0006**	0.0077	5.9E-155	**0.0679**	0.2578	9.0E-156
PFM_monalysin-like	**0.0137**	0.0221	2.0E-28	**0.0048**	0.0070	6.8E-13	**0.2691**	0.3241	1.2E-26
Mb-like	**0.0087**	0.0496	1.2E-159	**0.0022**	0.0219	4.8E-156	**0.1687**	0.4546	6.3E-160
FYVE_like_SF	**0.0114**	0.0449	1.6E-137	**0.0047**	0.0239	2.8E-134	**0.1701**	0.3165	5.8E-139
PFM_aerolysin_family	**0.0460**	0.0772	4.8E-152	**0.0103**	0.0316	9.3E-138	**0.6402**	0.7600	5.9E-118
7tmA_photoreceptors_insect	**0.0013**	0.0025	3.1E-69	**0.0002**	0.0010	3.3E-75	**0.0479**	0.0850	6.0E-75
7tmA_Melanopsin-like	**0.0009**	0.0039	2.1E-10	**0.0001**	0.0018	1.9E-10	**0.0430**	0.1168	2.1E-10
ClyA_AhlB-like	**0.0017**	0.0037	4.2E-49	**0.0002**	0.0008	1.2E-81	**0.0297**	0.0717	1.6E-84
7tmA_Opsins_type2_animals	**0.0020**	0.0083	6.8E-163	**0.0006**	0.0037	1.1E-154	**0.0700**	0.2326	6.0E-164
7tm_GPCRs	**0.0227**	0.0988	4.6E-24	**0.0053**	0.0713	5.6E-24	**0.5332**	0.8910	6.3E-24
7tmA_Anaphylatoxin_R-like	**0.0039**	0.0108	1.6E-23	**0.0012**	0.0037	3.8E-22	**0.1099**	0.2148	6.7E-24
7tmA_Opioid_R-like	**0.0020**	0.0071	7.8E-19	**0.0010**	0.0034	2.7E-17	**0.1175**	0.1939	3.5E-18
ClyA-like	**0.0233**	0.0619	1.2E-155	**0.0134**	0.0371	5.2E-145	**0.4284**	0.6702	1.1E-157
FGGY	**0.0045**	0.0110	2.1E-71	**0.0022**	0.0048	3.4E-45	**0.1888**	0.3539	7.6E-72
FGGY_YpCarbK_like	**0.0016**	0.0053	5.7E-78	**0.0002**	0.0015	4.5E-82	**0.0637**	0.1077	3.0E-89
7tm_classA_rhodopsin-like	**0.0058**	0.0261	5.1E-165	**0.0019**	0.0122	1.3E-161	**0.1844**	0.4528	5.2E-165
NBD_sugar-kinase_HSP70_actin	**0.0341**	0.1162	3.0E-161	**0.0178**	0.0983	1.2E-155	**0.6621**	0.8883	6.3E-164
7tmA_amine_R-like	**0.0049**	0.0215	3.6E-164	**0.0026**	0.0152	2.1E-155	**0.1917**	0.3560	5.4E-165

ProtT5-score outperforms BLOSUM45 greatly for long alignments

- The percentage of cases when ProtT5-score and BLOSUM45, resp., are producing better results, sorted increasingly by MSA length.
- For the much more difficult case of long alignments, ProtT5-score outperforms BLOSUM matrices in 94% of cases of alignments with length over 300, whereas BLOSUM45 is better in only 4% of those cases.



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